KRISHI-DRISHTI: AN AI-POWERED AGRICULTURAL INTELLIGENCE PLATFORM FOR INDIAN SMALLHOLDER FARMERS

Project Report

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1 INTRODUCTION

India is home to approximately 146 million farming households, the majority of which are smallholder farmers cultivating less than two hectares of land. These farmers, who together produce nearly 51 percent of India's food output, continue to operate with very limited access to real-time market information, modern agricultural advisory, government welfare schemes, and financial instruments such as crop insurance and carbon finance.

The agricultural sector in India contributes roughly 17 percent of the national GDP and employs more than 42 percent of the country's workforce. However, farmer income remains deeply inconsistent, partly due to information asymmetry, lack of direct market access, climate uncertainty, and the absence of digital tools that are designed specifically for the rural Indian context. Despite the proliferation of smartphones even in semi-rural areas, no existing platform consolidates the full range of services a farmer needs into a single, easy-to-use, Hindi-first mobile application.

Krishi-Drishti, which translates to "Farmer Vision" in Hindi, is a comprehensive AI-powered agricultural intelligence platform built to address precisely this gap. The platform integrates artificial intelligence, satellite remote sensing, machine learning, and real-time data APIs into a single unified interface available in both Hindi and English.

The platform serves as a digital companion for Indian farmers by providing:

- **AI-powered crop advisory** through a multilingual chat interface powered by Google Gemini.
- **A direct agricultural marketplace (Mandi)** where farmers can list produce and connect directly with buyers, bypassing exploitative middlemen.
- **Satellite-based crop health monitoring** using NDVI and EVI metrics derived from Google Earth Engine.
- **AI-powered plant disease diagnosis** using Google Gemini Vision to identify pathogens from photos.
- **A Carbon Vault module** to track and enroll sustainable farming practices into verified carbon credit pipelines.
- **Scheme Setu**, a portal for government welfare scheme discovery and digital application.
- **Real-time weather forecasting** with a 7-day outlook and disease-risk alerts.
- **Bioacoustic analysis** using the device microphone to detect pests or disease early via canopy vibration analysis.
- **An AI-powered Soil Carbon Model** estimating soil organic carbon from satellite-derived metrics.

2 LITERATURE SURVEY

2.1 Existing System

2.1.1 AgriStack and PM-KISAN Digital Infrastructure

The Indian government's AgriStack initiative builds foundational infrastructure including the Farmers' Database and Soil Health Cards. While beneficial, these are administrative systems and lack interactive advisory, market access, or disease diagnosis capabilities.

2.1.2 Agrimarket and eNAM

eNAM is a pan-India electronic trading portal. However, it requires significant smartphone proficiency, suffers from internet connectivity issues in mandis, and lacks AI advisory or carbon credit integration.

2.1.3 AgroStar and Kisan Suvidha

These offer agronomy advisory and input purchasing. However, they are primarily one-dimensional and do not integrate satellite monitoring or conversational AI powered by real remote sensing data.

2.1.4 Boomitra Carbon Platform

An aggregator-only model where farmers cannot choose their aggregator. It is English-primary and the onboarding process is lengthy ($\sim$30-45 minutes), making it inaccessible to many local users.

2.1.5 Grow Indigo

Targets large-scale organized farms with limited penetration among smallholders. It is not mobile-first and lacks real-time satellite-driven AI advisory.

2.1.6 EKI Energy Services

Uses manual, expensive third-party audits. The process takes months, and farmers have no direct visibility into their credit status or earnings.

2.1.7 Summary of Gaps in Existing Systems

- Lack of a single consolidated interface for advisory, market, satellite, and carbon credits.
- Absence of truly Hindi-first, mobile-first design for rural contexts.
- Opaque, slow, and intermediary-dependent carbon credit platforms.
- Satellite data inaccessible to individual smallholder farmers.

2.2 Proposed System

2.2.1 Multilingual AI Advisory

Conversational AI powered by Google Gemini, responding in Hindi and English to provide scientifically grounded advisory.

2.2.2 Direct-to-Farmer Digital Marketplace

A Mandi module allowing farmers to list produce directly, eliminating commission agents (“arhatiyas”).

2.2.3 Satellite-Based Farm Intelligence

Automated retrieval of NDVI, EVI, and MSAVI indices per plot polygon from Google Earth Engine.

2.2.4 AI Plant Disease Vision Diagnosis

Pathogen identification using Gemini Pro Vision from images of diseased plant tissue.

2.2.5 Carbon Credit Enrollment and Tracking

Transparent tracking of baseline emissions, sequestration, and available credits in a dashboard aligned with Verra VM0042 standards.

2.2.6 Scheme Setu

A government scheme aggregator tailored to farmer profiles (land size, district, crop type).

2.2.7 Disease Risk Forecasting

Epidemiological models utilizing Open-Meteo weather history to alert for powdery mildew or leaf blight risk.

2.2.8 Bioacoustic Pest Detection

Experimental signal processing of soil/canopy audio to identify pest infestation frequencies.

2.3 Problem Statement

- **P1 - Lack of Timely Market Information:** Losses of 20-40% due to information asymmetry.
- **P2 - Delayed Crop Disease Detection:** Yield losses of 10-50% due to slow identification.
- **P3 - Absence of Real-Time Farm Monitoring:** Lack of affordable satellite access for individuals.
- **P4 - Carbon Market Exclusion:** Complexity and literacy barriers prevent access to climate finance.
- **P5 - Poor Government Scheme Awareness:** Over 40% of eligible farmers are unaware of benefits.
- **P6 - Climate Vulnerability:** Lack of preventive advisory support for extreme weather.

2.4 Objectives

- **O1:** Implement a unified, mobile-first agricultural intelligence platform.
- **O2:** Integrate AI vision diagnosis via Google Gemini.
- **O3:** Develop a direct agricultural marketplace (Mandi).
- **O4:** Implement satellite-derived health monitoring (NDVI/EVI/MSAVI).
- **O5:** Build a transparent carbon credit enrollment and tracking module.
- **O6:** Integrate multilingual conversational AI in Hindi and English.
- **O7:** Develop the Scheme Setu application portal.

- **O8:** Implement ML-based disease risk forecasting.
- **O9:** Build an anomaly detection subsystem using Isolation Forest.
- **O10:** Demonstrate bioacoustic supplement for disease detection.

3 SYSTEM REQUIREMENTS

3.1 Hardware Requirements

3.1.1 Farmer-Side Hardware

- **Device:** Android 6.0+ or iOS 13+ with minimum 2GB RAM.
- **Peripherals:** GPS receiver, microphone, and 5MP+ rear camera.
- **Connectivity:** Minimum 3G (1 Mbps).

3.1.2 Server-Side Hardware

- **App Server:** 4-core CPU, 8GB RAM (AWS t3.large or equivalent).
- **Database:** PostgreSQL with PostGIS extension.
- **Storage:** AWS S3 for satellite cache and evidence photos.

3.1.3 IoT Hardware (Future Enhancement)

- **Sensors:** Capacitance-based soil moisture sensors (3-5 per plot).
- **Module:** LoRaWAN or 4G/NB-IoT uplink with solar power.

3.2 Software Requirements

- **Frontend Stack:** React 18, TypeScript 5, Vite 5, Lucide-React.
- **Backend Stack:** Python 3.11+, FastAPI, SQLAlchemy 2.x.
- **AI/ML Stack:** Google Gemini API, ee (Earth Engine), scikit-learn.
- **External APIs:** Open-Meteo, Razorpay.

4 SYSTEM DESIGN

4.1 Architecture / Workflow

4.1.1 Layered System Architecture

The system consists of five functional layers:

1. **Presentation Layer:** React frontend with 22 screens (Chat, Vision, Map, etc.).
2. **API Gateway Layer:** FastAPI endpoints organized into 14 domain-specific routers.
3. **Service Logic Layer:** Wrappers for Earth Engine, Anomaly Detection, and Forecasting.

4. **Data Layer:** 14 tables including users, plots, carbon_projects, and disease_alerts.
5. **Integration Layer:** Third-party calls to Gemini, S3, and Razorpay.

4.1.2 Authentication and Security Workflow

Uses phone-number-based OTP verification. On success, a JWT signed with RS256 is generated (30-min expiry), allowing secure access to protected routes.

4.1.3 Carbon Credit Calculation Methodology

- **Baseline:** Area $\times$ 3.2 (tCO$_2$e/ha/year).
- **Sequestration:** NDVI Value $\times$ Area $\times$ 1.8.
- **Issuance:** (Sequestration $-$ 15% Buffer) $\times$ (1 $-$ 20% Platform Fee).

4.1.4 Machine Learning Components

- **NDVI Anomaly Detection:** Uses **Isolation Forest** on historical time series data.
- **Disease Risk Forecasting:** Epidemiological model calibrated against ICAR thresholds.
- **Soil Carbon Estimation:** Trailing average NDVI regressed against soil texture and rain.

4.1.5 Real-Time Satellite Analysis Workflow

When a farmer selects a plot, the backend reads the GeoJSON polygon and uses the Earth Engine Python SDK to query Sentinel-2 collections (cloud cover $<$ 20%). NDVI is computed as $(NIR - RED)/(NIR + RED)$.

5 IMPLEMENTATION

5.1 Proposed Methodology

5.1.1 Technology Architecture

Independent deployable units: React PWA (Frontend) and FastAPI (Backend). Navigation state is managed via a central machine in `App.tsx`. Backend persistence uses SQLAlchemy with an environment-driven database URL.

5.1.2 Modular Implementation: Priority Order

1. **Phase A (Foundation):** DB Schema, Auth (JWT/OTP), Navigation Shell.
2. **Phase B (Core):** Marketplace, AI Chat, Vision, Weather, Map modules.
3. **Phase C (Advanced):** Carbon Vault, Scheme Setu, Insurance, Community.
4. **Phase D (AI/ML):** Anomaly detection, Soil carbon model, Bioacoustic scanner.

5.1.3 Carbon Vault Implementation

Divided into Enrollment (baseline), Evidence Collection (geotagged practice photos), and Verification (NDVI delta analysis). Payouts are automated via Razorpay after a 90-day verification cycle.

5.1.4 Disease Risk Forecasting Design

Features include Min/Max Temp, Humidity, and Rainfall. Alerts specify the risk level (High/Medium/Low) and localized recommendations based on ICAR management protocols.

5.1.5 Testing and Deployment

Frontend uses Vitest/React Testing Library. Backend uses `pytest` with `httpx` for async tests. Target line coverage for the service layer is 80%. Production deployment utilizes Docker on AWS EC2 with multi-AZ RDS PostgreSQL.

6 CONCLUSION AND FUTURE WORK

6.1 Achievements

- Successful integration of 10 functionally distinct AI/Satellite modules.
- Low-cost verification model (86% confidence) for smallholders.
- Transparent carbon credit pipeline and direct Mandi access.

6.2 Future Work

- **FW1:** Blockchain-based credit registry (Polygon ERC-20 tokens).
- **FW2:** LoRaWAN IoT sensor integration for direct ground truth.
- **FW4:** Offline-first mobile app using SQLite caching.
- **FW7:** Voice interface for low-literacy users.

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